Rahul P A

me25mtech11022@iith.ac.in | rahulpa369@gmail.com | +91 9207203087 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.36
St. Joseph's Central School, Mundakayam	Class XII (CBSE)	2018	86.80%
St. Joseph's Central School, Mundakayam	Class X (CBSE)	2016	10.00%

SCHOLASTIC ACHIEVEMENTS

Secured 9th position in B.Tech Mechanical Engineering Exam, Cochin University of Science and Technology, Kochi.

SKILLS

CFD and CAE: ANSYS Fluent, OpenFOAM, ANSYS ICEM CFD, Gmsh, ParaView

CAD: SolidWorks, AutoCAD, Fusion 360, FreeCAD

Programming: C, C++, Python, MATLAB, SQL

Other: Microsoft Office, Figma

PROJECTS

Numerical Investigation of Start/Unstart Phenomena in a Streamline-Traced Hypersonic Busemann Inlet | M.Tech Thesis, OpenFOAM

- **Developed** an in-house Busemann contour generator solving the Taylor-Maccoll equations (RK4 with double-shooting iteration), reproducing three independently published designs within 2% and most within 0.5%, and extended it to streamline-traced sugarscoop geometry via cos- tracing.
- **Verified** rhoCentralFoam against a DLR TAU benchmark at Mach 8 across four wall-resolved k- SST meshes (21k-2.38M cells, y+ < 1), matching reference wall pressure within 0.6% at 0.23% grid convergence.
- **Traced** a 42% over-compression above the inviscid design pressure ratio to boundary-layer displacement and shock-cancellation failure, justifying axisymmetric wedge meshes over full 3D at ~100× lower computational cost.
- **Simulating** 3D compressible flow to resolve shock-train structure, shock/boundary-layer interaction and unstart onset, with the flowfield validated against experimental data at Mach 6.
- **Conducting** a parametric study across design and off-design conditions to map shock-train evolution and quantify the performance limits that govern inlet unstart.

Assessment of RANS Predictive Capability for Hypersonic Shock-Wave / Turbulent Boundary-Layer Interaction | ANSYS Fluent, C

- **Engineered** seven compiled Fluent User-Defined Functions, including a User-Defined Scalar transport equation for shock detection, to implement shock-unsteadiness and variable turbulent Prandtl number closures unavailable natively in the solver.
- **Reduced** peak reattachment heat-flux error from approximately 300% to 16% and separation-length error from -75% to +9% against Schulein's Mach 5, 14-degree oblique impinging-shock experiment.
- **Stabilised** a divergent scalar-turbulence coupling through implicit source linearisation and field smoothing, achieving grid independence on a 125,000-cell structured mesh at under 1% variation in peak heat flux.

Shock-Free Supersonic Nozzle Design by Method of Characteristics | MATLAB, ANSYS Fluent

- **Designed** a Mach 2.157 shock-free planar nozzle contour in MATLAB using 20 characteristic lines with Prandtl-Meyer and compatibility relations, matching isentropic area-ratio theory exactly.
- **Meshed** the converging-diverging domain into 24,000 structured quad cells with 0.12 max skewness and 0.93 min orthogonal quality.
- **Validated** the design in Fluent using the density-based inviscid solver with Roe-FDS flux splitting, converging all residuals below 1e-6 and recovering exit Mach 2.120 within 1.71% of the analytical design point.

Prediction of Mechanical Properties of Iron Alloys | Python, scikit-learn

- **Built** an end-to-end multi-output regression pipeline predicting Ultimate Tensile Strength and Elastic Modulus from alloy composition across 1,000 samples, using regex-based numeric extraction, metallurgy-informed feature engineering and imputation to expand the input space to 97 features.
- **Benchmarked** five model families (Random Forest, Extra Trees, Gradient Boosting, Ridge, ElasticNet) under 5-fold cross-validation and combined the top three in a voting ensemble, achieving R2 of 0.9927 for UTS and 0.9074 for Elastic Modulus.

Material Suitability Prediction for Electric Vehicle Chassis | Python

- **Formulated** material suitability rating as an ordinal regression problem rather than classification, justified by the ordered rating scale and severe class imbalance in the dataset.
- **Developed** and compared Ridge Regression against an Artificial Neural Network on experimentally validated material property data, with the ANN achieving R2 of 0.9631 and RMSE of 0.2963.

CNN-Based Image Classification on CIFAR-10 | PyTorch

- **Designed** a baseline CNN from scratch with batch normalisation, progressive dropout and data augmentation, reaching 73.2% test accuracy across ten balanced classes.
- **Tuned** learning rate, batch size and optimiser through a subset-based sweep, then adapted ResNet-18 for 32x32 input resolution, raising test accuracy to 87.8% at a macro F1-score of 0.876.

Overhauling and Performance Characterisation of a Francis Turbine Test Rig | B.Tech Project

- **Diagnosed** the efficiency loss in an 18 m head, 1500 rpm laboratory Francis turbine through full disassembly and inspection, tracing it to a worn bush and failed water seals that admitted water into the deep-groove bearing, compounded by leakage and pipeline fouling.
- **Restored** the rig over a 171-day overhaul by machining a gun-metal bush, replacing the 6308 and 1308 bearings, seals, gland packing and instrumentation, and correcting shaft alignment before recommissioning, delivered within a 40,000 budget.
- **Characterised** the rebuilt turbine using venturi flow measurement and rope-brake dynamometry, raising peak overall efficiency from 57% to 77% and returning the rig to instructional use.

Design of a 15 kW Single-Cylinder Air-Cooled Diesel Engine | B.Tech Mini Project, SolidWorks

- **Sized** a 78 × 97.5 mm bore-stroke (466 cc) engine for 15 kW at 4,500 rpm, verifying cylinder stresses within a factor-of-safety-6 allowable.
- **Dimensioned** the piston, I-section connecting rod (Rankine buckling) and crankshaft, holding crank-web stress to 29.3 MPa against a 145.6 MPa limit.
- **Modelled** the full assembly in SolidWorks with a 12-fin aluminium stack rejecting 2.96 kW, eliminating a liquid cooling circuit.

Multi-Task Deep Learning on Engine Sensor Time-Series | TensorFlow / Keras

- **Developed** a cascaded CNN-LSTM pipeline for simultaneous classification of engine operational mode and regression of fuel efficiency from time-stamped sensor data (temperature, RPM, vibration, torque, power).
- **Enforced** a strictly sequential 80/20 split with sliding-window sequencing to eliminate look-ahead data leakage, and diagnosed the dataset as noise-dominated when regularised models converged to mean-prediction behaviour rather than overfitting.

RELEVANT COURSES

MTech Courses: Computational Fluid Dynamics, Incompressible Fluid Flow, Turbulence, Advanced Heat Transfer, Compressible Flow and its Computation, Advanced CFD, Machine Learning and its Applications

BTech Courses: Fluid Mechanics, Heat Transfer, Thermodynamics, Strength of Materials, Machine Design, Manufacturing

Processes, Refrigeration and Air Conditioning, Mechatronics, Power Plant Engineering, Thermal Engineering, CAD/CAM

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Department of Mechanical & Aerospace Engineering, IIT Hyderabad

- **Heat Transfer Lab (ME4445)** and **Fluid Mechanics Lab (ME2431)** -- B.Tech students
- Conducted lab experiments and tutorial sessions for B.Tech cohorts, held doubt-clearing sessions, evaluated lab records, and maintained experimental setups and lab equipment.

EXPERIENCE

Managed Network Expert -- Mechanical Engineering | Chegg India Pvt. Ltd. (Part-time)

- Authored step-by-step solutions to undergraduate mechanical engineering problems spanning thermodynamics, fluid mechanics, heat transfer and machine design as an engaged subject expert on Chegg's Q&A platform.

Industrial Internship | HMT Machine Tools Ltd. (Govt. of India Undertaking), Kalamassery (2 weeks)

- Completed a structured plant-wide familiarisation covering foundry, machine shop, assembly and quality departments, studying the end-to-end manufacturing route for machine tool production.

EXTRACURRICULAR

Student Member, Indian Green Building Council (IGBC), CUSAT Chapter

ArduBotics -- National Level Workshop on Arduino-Based Robotics, Technex'20 (IIT BHU) outreach, held at CUSAT

Google Cloud Ready Facilitator Program

UI/UX Design Bootcamp (7 days) -- TinkerHub KMCTCE and TIST Cochin

SEMINARS

Developing Heat Transfer in Rectangular Channels with Rib-Turbulators | M.Tech Seminar, IIT Hyderabad

- Reviewed experimental and numerical literature on turbulated internal cooling channels, examining how rib geometry, pitch-to-height ratio and angle of attack govern the trade-off between heat transfer enhancement and pressure-drop penalty in the developing-flow region.

Micro-Electro-Mechanical Systems (MEMS) | B.Tech Seminar, Cochin University of Science and Technology

- Surveyed MEMS device architecture and micro-fabrication, covering the working principles of pressure, inertial and optical MEMS devices -- diaphragm-based pressure sensing, spring-mass accelerometers, vibrating-structure gyroscopes and digital micromirror arrays -- and their deployment across automotive safety, biomedical instrumentation and display technology.

CERTIFICATIONS AND TRAINING

Google-Supported Virtual Internship Program, "Android Basics in Kotlin" -- SmartInternz, AICTE

Machine Learning Summer Training - Analytics Vidhya

JavaScript Algorithms and Data Structures Developer Certification -- freeCodeCamp (~300 hours)

AutoCAD Mechanical Training -- LearnVern

MATLAB Onramp -- MathWorks

The Fundamentals of Digital Marketing -- Google Digital Unlocked

The C++ Workshop -- Packt

HOBBIES AND INTERESTS

Badminton, Yoga, Malayalam cinema, Creating educational short-form video content